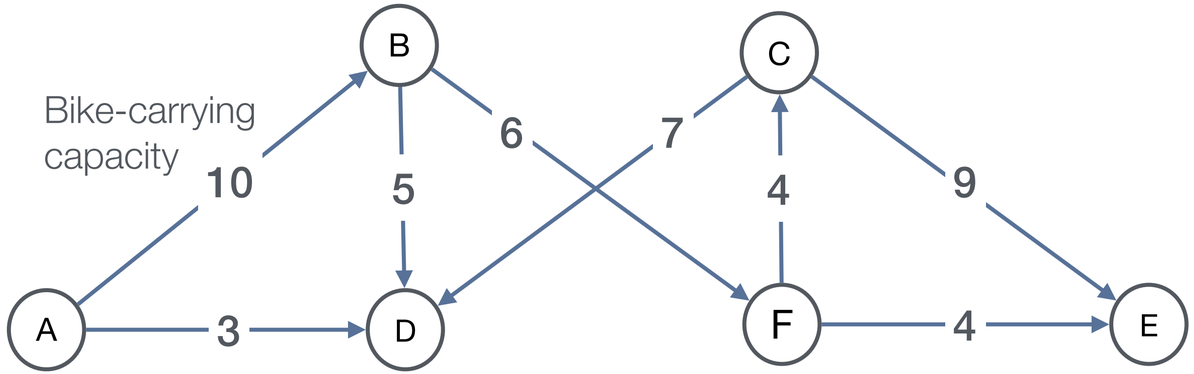
In-Class Assignment L5-2 (Group of Three or Less): Bike Sharing



(25 minutes) Submit to LEB2

A bike sharing company has six locations for check-out and return. At the end of each day, the number of bikes reserved for the next day and the number of bikes available in each location are determined and given below. A van with certain bike-carrying capacities is used to transport bikes among locations at night to make sure that each location has sufficient bikes in the next morning. The bike-carrying capacities among locations are shown in the graph below.



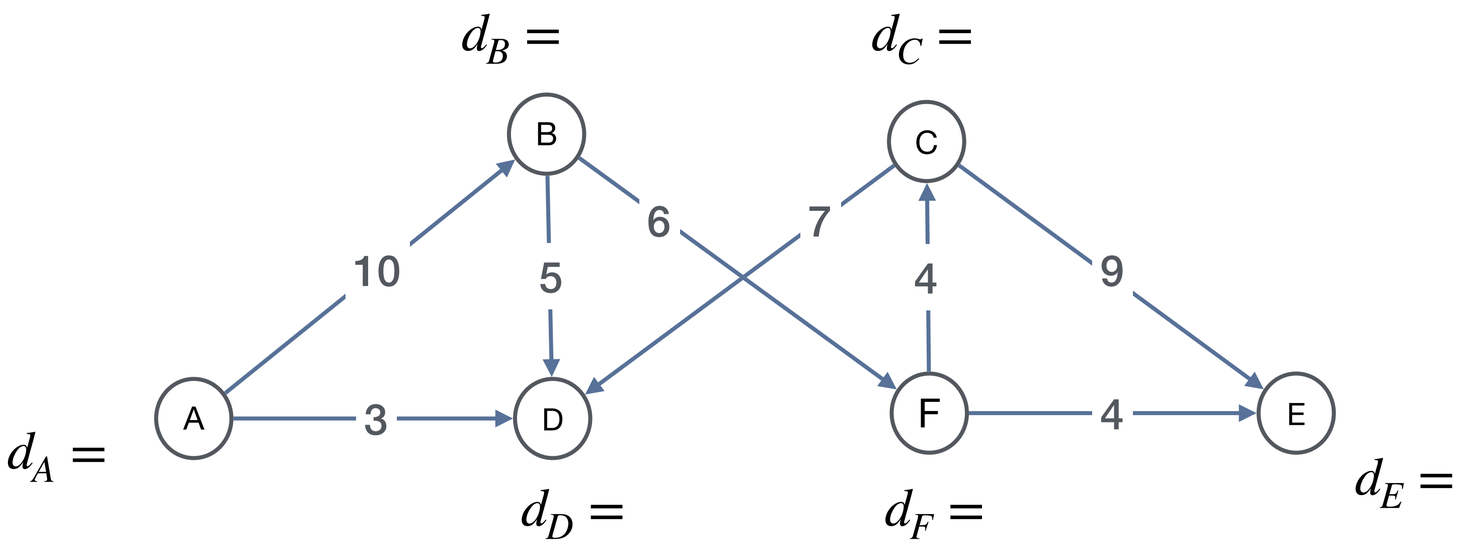
Location	Reserved	Available
A	9	17
B	12	20
C	12	18
D	20	10
E	26	15
F	0	0

From the given bike data and the bike-carrying capacity graph,

(a) Determine the supply and demand values at each location based on the number of available bikes and reservations.

(b) Model the problem as the max-flow problem by drawing the corresponding flow network.

(a) Supply and demand values at each location



(b) Corresponding flow network of the problem

